

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2370

Garden 138 kV
November 15, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2370. The Interconnection Customer proposes a 240.5 MW solar photovoltaic facility interconnecting at the Garden 138 kV substation in the Virginia area. The request was received on December 31, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 571 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,909,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Gaston 345-138 kV TX is overloaded to 136.2% of its 327 MVA emergency rating for the loss of the Garden to Dpc Elroy 138 kV. The project contributes 81.4% DFAX (195.7 MW) to this overload.

The Racine to Lawrenceville 138 kV Line is overloaded to 112.5% of its 426 MVA emergency rating for the loss of the Garden to Dpc Elroy 138 kV. The project contributes 99.9% DFAX (240.2 MW) to this overload.

For the loss of Lawrenceville to Racine 138 kV, the Dpc Elroy to Garden 138 kV line loads to 109.5% of its 438 MVA rating. Project DFAX: 99.9%.

For the loss of Garden Substation to Dpc Elroy 138 kV, the Racine to Gaston 138 kV loads to 104.9% of its 441 MVA rating. Project DFAX: 99.9%.

For the loss of Switzer to Pheasant Branch 138 kV Line, the Potosi to Tyler 138 kV loads to 100.1% of its 396 MVA rating. Project DFAX: 5.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gaston 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$17,439,000	55% (\$9,654,000)
Racine to Lawrenceville 138 kV Line	Rebuild line with higher-capacity conductor	\$5,317,000	57% (\$3,057,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Garden to Dpc Elroy 138 kV	Reconductor line	\$5,657,000	57% (\$3,252,000)
Racine to Gaston 138 kV	Reconductor line	\$3,362,000	57% (\$1,933,000)
Potosi to Tyler 138 kV	Reconductor line	\$214,000	6% (\$13,000)
	Total allocated to project		\$17,909,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.